AGENTHON 2026 | TRACK 2 FORECASTING | V6

T2 Quant V6

Final Integration Model & Submission Pipeline Report

Core question | Can we freeze only the components that survived V1–V5 validation and build a conservative final candidate that also runs reliably in the official submission environment?

PCA2 native joint + optional V4.3 text-tail overlay
Daily V6 + Monthly validated fallback + House-model safe fallback

Development Report | 2026-10-02

1. Position of V6 – integration, not another complex model

V6 does not introduce another standalone forecasting engine. It freezes the components that survived V1–V5 and combines them into a conservative final candidate with a production-safe submission path.

Version	Main experiment	Decision
V1	EWMA + bootstrap + weak drift	Marginal/Tail improved; Joint deteriorated
V2	Student-t dependence	Did not solve the Joint bottleneck
V3	PCA factor model	PCA2 improved over block1 but still lagged Reference Joint
V4	Text-conditioned tail mixture	Text can adjust tails; V4.3 shrank the range for robustness
V5	Copula reorder / shrinkage / adaptive gate	Promising development results failed fresh holdout
V6	PCA2 + optional V4.3 + production router	Freeze validated components and integrate submission path

Core V6 decision: reject the V5 dependence-overwrite branches and preserve PCA2 native joint structure. Text does not directly choose a price forecast; it only adjusts a conservative tail-mixture overlay.

2. Final architecture

Input	Router	Daily V6	Text condition	Output
panel + card + text	check target_frequency	PCA2 (2 factors, r=0.10, recent=0.60)	no signal/failure → PCA2 signal → V4.3 5–10% tail overlay	joint Monte Carlo draws
monthly card	→ validated monthly reference	preserve publication-lag / horizon logic	text not used	same submission schema

Daily cards use V6. With no usable text signal, or when the House call fails, the output is exactly the frozen PCA2 distribution. With a validated structured signal, V4.3 adds only a stress-mixture overlay on the same PCA2 path engine.

Monthly cards are not forced through the daily PCA engine. They retain the validated reference monthly implementation with publication-lag and observation-period handling.

Final daily distribution:

$$F_{V6} = (1 - p_{tail})\, F_{PCA2} + p_{tail}\, F_{stress}$$

When text_signal is None, the overlay is bypassed, so $F_{V6} = F_{PCA2}$ exactly.

3. Numeric backbone – frozen PCA2 specification

No further parameter tuning is performed after V3.1. The daily V6 backbone is frozen as follows.

Component	V6 setting	Interpretation
PCA factors	2	Prefer a low-dimensional Level/Slope-like structure
Residual scale	0.10	Allow limited idiosyncratic noise
Factor recent mix	0.60	Emphasize recent regime while retaining long history
EWMA half-life	60	Give greater weight to recent volatility
Target	joint Monte Carlo distribution	Forecast the full joint distribution, not a point estimate

On the representative 24-fold UST benchmark, PCA2 r10/m60 beat the Reference on Marginal and Tail but still lagged on Joint, leaving Composite about 1.5% worse.

Metric ↓	Reference	V6 numeric (PCA2)	Relative change
Marginal	1.1858	1.1409	~3.8% better
Joint	4.4381	4.6291	~4.3% worse
Tail	0.1579	0.1291	~18.2% better
Composite	1.9559	1.9850	~1.5% worse

4. Text layer – V4.3 conservative tail gate

The text layer does not generate an expected return, covariance matrix, or scenario path. The House Model returns only five structured fields, and Python maps them deterministically into a tail probability.

Field	Meaning
shock_direction	upper / lower / both / none in target units
shock_severity	shock magnitude [0,1]
confidence	directness and consistency of textual evidence [0,1]
immediacy	relevance to the forecast horizon [0,1]
regime	short descriptive regime label

$$\text{activation} = \text{severity} \times \text{confidence} \times \text{immediacy}$$

$$p_{\text{tail}} = 0.05 + (0.10 - 0.05) \times \text{activation}$$

V4.3 narrows the earlier 5–15% gate to 5–10%. The objective is not the maximum local score, but lower damage from over-activation such as the downgrade-watch failure case.

Implementation nuance: signal=None means pure PCA2. A supplied signal with direction="none" still activates the 5% dormant stress mixture. Provider failures therefore fall back to None rather than fabricating a neutral signal.

5. Leakage control and House-model adapter

Leakage control is the central constraint for text. The adapter reads corpus_index.json and includes only documents with timestamp ≤ as-of.

Step	Action
1. Corpus load	Read corpus_index.json and referenced text files
2. Cutoff filter	Exclude documents after the forecast origin
3. Prompt	No outside knowledge/web/tools; request JSON-only signal
4. House call	Use MODEL_ENDPOINT / MODEL_TOKEN / MODEL_NAME; one request; no automatic retry
5. Parse/validate	Validate schema and all [0,1] fields
6. Failure policy	Missing environment, network/API/parse failure → signal=None → pure PCA2

Fail-safe principle: failure of the text layer must not fail the entire forecast unit. The numeric backbone remains independently admissible.

6. Why V5 joint corrections were rejected

V5 produced the most important negative result for the final design. Reference-copula rank reordering improved one election unit, but the effect did not generalize across fresh units.

Experiment	Development impression	Fresh/holdout result	Decision
V5.0 full reference copula	Strong gain on one UST unit	6-unit mean +3.05% worse vs PCA2	Reject
V5.1 λ=0.25 rank shrinkage	Development 6-unit mean -0.26% better	fresh 8-unit mean +0.59% worse	Reject
V5.2 past-performance gate	Regime-specific selector idea	test mean +0.94% worse vs PCA2	Reject

This does not mean regime-dependent model selection is conceptually wrong. It means that the selector must predict which model will win before seeing future realizations. V5.2 showed that past model rankings were not persistent enough across regimes.

V6 therefore fixes one numeric backbone and permits only a small text-conditioned tail adjustment, reducing model-selection overfitting.

7. How to interpret current performance

There is not yet one defensible number such as “V6 beats Reference by X% overall.” Public final outcomes are sealed, and the final House-text V6 branch has not been rerun under one identical cross-unit benchmark with realized targets.

Evidence	Result	Interpretation
UST 24-fold	PCA2 composite 1.9850 vs Ref 1.9559	~1.5% worse; Joint remains the bottleneck
UST components	Marginal ~3.8% better, Tail ~18.2% better	Strong calibration/tail advantage
CPI text-covered 5 folds (V4.2b)	0.2463 vs PCA2 0.2484 vs Ref 0.2504	Text gate helped in a small sample
V4.3 cross-unit vs fixed 10%	fold-weighted ~-0.40%, unit-equal ~-0.13%	Robustness favored over aggressive activation
V5 holdouts	Joint corrections worsened on average	Keep V5 branches out of final model

Current position: V6 cannot be claimed to dominate the Reference consistently. It has structural advantages in Marginal and Tail, and it wins by large margins in some cross-asset regimes, but rates Joint coherence can still favor the Reference. The final design prioritizes generalization and submission robustness over local-score maximization.

8. End-to-end validation

V6 was validated not only as model code, but through the actual submission executable and a Linux Docker environment.

Validation	Result
V6 no-text vs direct PCA2 smoke	max absolute difference = 0
V6 text vs direct V4.3 smoke	max absolute difference = 0
Local CLI – no House env	PCA2 fallback; g0–g3 all pass
Local CLI – manual text signal	p_tail=0.0598; g0–g3 all pass
Production forecast entrypoint	production_cli routing verified
Monthly regression unit	reference monthly fallback; g0–g3 all pass
Docker build	python:3.13-slim-bookworm image built successfully
Docker Linux end-to-end	V6 text branch + exactly 3 output files + g0–g3 all pass

The Docker test used an /input read-only and /output writable layout with network=none and a manual structured signal. The output directory contained exactly forecast.parquet, forecast_meta.json, and forecast_rationale.md.

9. Production routing and submission robustness

Both pyproject.toml and Dockerfile route the forecast verb to production_cli. The dispatcher reads card.toml and selects a branch from target_frequency.

Condition	Execution branch	Reason
daily + text signal	PCA2 + V4.3	Apply text-conditioned tail risk
daily + no signal / House failure	pure PCA2	Fail-safe numeric forecast
monthly	validated reference monthly CLI	Preserve publication-lag and observation-period handling
manual local test	--text-signal-json	Reproduce the text branch without House API access

The only unverified external dependency is a live call to the competition House endpoint. The adapter deliberately avoids vendor-specific hard-coding and uses the official MODEL_* environment contract with an OpenAI-compatible /v1/chat/completions path.

10. V6 final checklist and next research direction

Question	V6 answer
Final daily backbone?	PCA2: 2 factors, residual 0.10, recent mix 0.60
What does text change?	Only tail-mixture probability and direction
House failure?	signal=None → pure PCA2
V5 joint correction?	Excluded after holdout failure
Monthly cards?	Keep validated reference monthly path
Local gates?	g0-g3 all pass
Docker/Linux gates?	g0-g3 all pass
Final public score?	Not available because realized outcomes are sealed

Next research candidate – V7 router / ensemble

Rather than adding another copula or factor engine, the natural next experiment is an ex-ante router/ensemble that uses observable regime features – PCA explained variance, correlation stability, EWMA/long-run volatility ratio, cross-sectional dispersion, and text activation – to set the weight between PCA2 and the Reference.

V6 is intentionally not the most complex model. It is the stable final candidate composed only of elements that survived validation. Future complexity should be accepted only when it improves fresh holdout performance relative to V6.